TechNova Software Hackathon

Team Name : Codexx

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Student Project Repository and Showcase Platform

Problem Statement

Many educational institutions still rely on manual methods or disconnected systems to manage student projects, making project submission, approval, storage, and retrieval inefficient. Students often face difficulties in showcasing their work, collaborating with peers, and accessing quality project resources. Faculty members spend significant time managing project reviews manually, while valuable student projects remain underutilized after completion. Additionally, the absence of a structured recognition system reduces student motivation to contribute innovative ideas. Therefore, there is a need for a centralized, secure, and collaborative platform that efficiently manages projects while encouraging knowledge sharing, innovation, and active student participation.

Existing System

In many educational institutions, student project management is carried out through manual processes or fragmented digital systems. Project submissions, faculty reviews, approvals, and document storage are often managed using emails, spreadsheets, or physical records, making the process time-consuming and difficult to track. There is no centralized repository for preserving previous projects, limited opportunities for collaboration and knowledge sharing, and no structured mechanism to recognize student contributions, resulting in reduced transparency, accessibility, and overall efficiency.

Proposed Solution

The proposed Student Project Repository and Showcase Platform provides a centralized web-based solution for managing Internal Projects, External Projects, and Project Ideas. The platform automates project submission, faculty assignment, review, approval, and repository management, reducing manual effort and improving transparency. It introduces a Credit Point System, student networking, leaderboards, notifications, and project collaboration features to encourage continuous participation and knowledge sharing. Repository access is based on student contributions, ensuring responsible use of academic resources. By integrating project management, collaboration, and recognition into a single platform, the system enhances learning, innovation, and overall academic productivity within the institution.

Project Overview

The Student Project Repository and Showcase Platform is a web-based application developed to provide a centralized environment for managing, storing, approving, collaborating on, and showcasing student projects. The platform simplifies the complete project lifecycle by replacing manual documentation and approval processes with a secure digital system. Students can submit Internal Academic Projects, External Projects, and Project Ideas, while faculty members review and approve project submissions to ensure quality and academic standards. Internal projects are automatically assigned to faculty members based on their domain expertise and availability, whereas students can select faculty guides for external projects. The platform also includes a Credit Point System that rewards students for approved projects, innovative ideas, project enhancements, and active participation. Repository access is controlled using contribution-based criteria, encouraging continuous engagement and knowledge sharing. Additional features such as student networking, notifications, leader-boards, project collaboration, and project engagement create an interactive academic ecosystem that promotes innovation, transparency, technical learning, and professional development within the institution.

Objectives

The primary objectives of the Student Project Repository and Showcase Platform are:

- To digitize the complete project submission, review, approval, and repository management process.
- To provide a centralized platform for storing, organizing, and showcasing Internal Projects, External Projects, and Project Ideas.
- To reduce manual paperwork and improve the efficiency of project management within the institution.
- To automatically assign faculty reviewers for internal projects based on project domain and faculty availability.
- To enable students to select faculty guides for external projects according to their project domain or department.
- To encourage creativity and innovation by providing an Idea Showcase platform for students to present their project concepts.
- To motivate students through a structured Credit Point System that rewards quality projects, innovative ideas, and collaborative contributions.
- To implement a secure repository access mechanism based on approved projects and earned credit points.

- To promote collaborative learning by allowing students to enhance and contribute to existing projects.
- To facilitate academic networking by enabling students to connect with peers and receive updates on their activities.
- To provide real-time notifications for project approvals, showcased projects, connection requests, and project submission deadlines.
- To increase student engagement through project likes, comments, views, leader-boards, and achievement recognition.
- To ensure transparency, fairness, and accountability in the faculty review and project approval process.
- To create a searchable knowledge repository that preserves valuable academic projects for future reference and learning.
- To support continuous technical learning, skill development, and professional growth among students through project showcasing and collaboration.

Purpose

The purpose of the Student Project Repository and Showcase Platform is to provide a centralized and secure platform for managing the complete lifecycle of student projects. It aims to simplify project submission, faculty review, approval, and repository management while promoting collaboration, innovation, and knowledge sharing among students. The platform encourages active participation through a Credit Point System, enhances project visibility, ensures transparent evaluation, and preserves valuable academic work for future reference, creating an efficient and sustainable digital project management ecosystem.

Scope

The system supports:

- Student Registration
- Authentication
- Internal Project Submission
- External Project Showcase
- Idea Showcase
- Faculty Assignment
- Faculty Approval
- Credit Point Management
- Repository Access Control
- Student Networking
- Project Collaboration
- Notifications
- Leaderboard
- Admin Management

Technology Stack

Component	Technology
Frontend	React.js
Backend	Node.js + Express.js
Database	PostgreSQL
Authentication	JWT
File Storage	Local Storage / Cloud Storage
Version Control	Git & GitHub

System Requirements

Software Requirements

Component	Specification
Operating System	Windows 10/11, Linux
Browser	Chrome, Edge, Firefox
Database	PostgreSQL
Backend	Node.js
Frontend	React.js
IDE	VS Code

Hardware Requirements

Component	Specification
Processor	Intel i5 or above
RAM	8 GB
Storage	256 GB SSD
Internet	Required

User Roles

Student

- Register/Login
- Upload Internal Projects
- Upload External Projects
- Upload Project Ideas
- View Repository
- Enhance Projects
- Earn Credit Points
- View Leaderboard

Faculty

- Review Internal Projects
- Accept External Guide Requests
- Review External Projects
- Approve/Reject Projects
- Award Credit Points
- Provide Feedback

Administrator

- View All Projects
- Remove Projects
- View Leaderboard
- Manage Users
- Monitor Repository

Functional Modules

Module 1 – User Authentication & Role Management

Description

Provides secure authentication and role-based access.

Features

- Student Registration
- Faculty Login
- Admin Login
- Logout
- Password Reset
- JWT Authentication
- Role-Based Dashboard

Module 2 – Student Profile Management

Description

Maintains complete student information.

Features

- Personal Information
- Department
- Academic Year
- Skills
- Domain of Interest
- Total Credit Points
- Approved Project Count
- Portfolio

Module 3 – Internal Project Management

Description

Handles academic projects completed as part of the curriculum.

Submission Includes

- Project Title
- Abstract
- Documentation
- GitHub Repository
- PPT
- Demo Video (Optional)
- Vercel Link (Optional)

Workflow

Student Uploads Project

↓

System identifies project domain

↓

Faculty automatically assigned based on:

- Domain Expertise
- Current Availability

↓

Faculty Reviews

↓

Approved Project Published

↓

Notifications

↓

Credit points updated

Faculty approval is mandatory.

No guide request is required.

Module 4 – External Project Management

Description

Allows students to showcase projects completed outside academics.

Examples

- Hackathons
- Freelancing
- Personal Projects
- Open Source
- Competitions

Submission Includes

- Documentation
- GitHub Repository
- Certificate (Optional)
- Demo Video (Optional)
- Vercel Link (Optional)

Workflow

Student selects project domain

↓

System displays eligible faculty

↓

Student sends guide request

↓

Faculty receives notification

↓

Faculty accepts/rejects

↓

Student uploads project

↓

Faculty reviews

↓

Repository Publication

↓

Credit points updated

Faculty approval is mandatory.

Module 5 – Idea Showcase Module

Description

Allows students to publish project ideas before implementation.

Submission Includes

- Idea Title
- Documentation
- GitHub Repository (Optional)
- Demo Video (Optional)
- Vercel Link (Optional)

Features

- Direct Publication
- No Faculty Approval Required
- Students earn Idea Credit Points

Module 6 – Faculty Assignment & Guide Management

Description

Manages faculty allocation.

Internal Projects

- Automatic faculty assignment
- Based on domain
- Based on faculty availability

External Projects

- Student selects preferred faculty
- Faculty accepts/rejects guide request

Module 7 – Credit Point Management

Description

Maintains the contribution score of every student.

Credit Allocation

Activity	Credits
Internal Project Approved	10
External Project Approved	20
Idea Published	5
Successful Project Enhancement (Owner)	10

Activity	Credits
10 Likes on Project	1

Credits are awarded by faculty after project approval.

Module 8 – Project Collaboration & Enhancement

Description

Allows students to extend or improve existing approved projects.

Workflow

Student requests project access

↓

Eligibility verified

↓

Project accessed

↓

Student enhances project

↓

Faculty verifies enhancement

↓

Project Owner receives Credit Points

The original owner receives credits for encouraging collaboration.

Module 9 – Repository Access Control

Description

Controls access to repositories based on student contribution.

Eligibility Criteria

Students must have at least 3 approved projects.

Additionally:

Credit Points	Access Level
< 60	No Repository Access
≥ 60	View Idea Repository
≥ 100	View Internal Projects
≥ 200	View External Projects

Students who do not satisfy the above criteria cannot access project repositories.

Module 10 – Project Repository & Search

Features

Browse

- Ideas
- Internal Projects
- External Projects

Search By

- Project Title
- Student Name
- Department
- Domain
- Technology

Sort By

- Latest
- Most Viewed
- Highest Credits
- Most Popular

Module 11 – Student Leaderboard

Description

Displays rankings of students according to total Credit Points.

Visible To

- Students
- Faculty
- Administrator

Ranking Criteria

- Total Credit Points
- Number of Approved Projects

Filters

- Overall
- Department
- Academic Year

Module 12 – Project Engagement Module

Description

Tracks project popularity similar to LinkedIn.

Features

- Likes
- Comments
- Views

Credit Policy

- Every 10 Likes = 1 Credit
- Comments do not generate credits
- Views do not generate credits

Engagement statistics improve project visibility.

Module 13 – Notification Module

Student Notifications

- Faculty Assigned
- Faculty Accepted Guide Request
- Project Approved
- Project Rejected

- Credit Points Awarded
- Collaboration Updates

Faculty Notifications

- New Guide Request
- New Project Submission
- Enhancement Verification

Admin Notifications

- New Project Uploaded
- Reported Projects
- Repository Updates

Module 14 – Admin Management Module

Description

Provides complete administrative control over the repository.

Features

- View All Projects
- Remove Projects
- View Leaderboard
- View Student Profiles
- Manage Repository
- Manage Departments
- View Project Statistics

Credit Point Policy

Activity	Credits
Internal Project Approved	10
External Project Approved	20
Idea Published	5
Project Enhancement (Owner Reward)	10
Every 10 Likes	1
Comments	0
Views	0

System Architecture

The Student Project Repository and Showcase Platform follows a Three-Tier Architecture, consisting of the Presentation Layer, Application Layer, and Database Layer. The Presentation Layer, developed using React.js, provides an interactive user interface for students, faculty, and administrators. The Application Layer, built with Node.js and Express.js, handles business logic, user authentication, project management, faculty assignment, credit calculation, notifications, and API communication. The Database Layer, implemented using PostgreSQL, securely stores user information, project details, reviews, credit points, notifications, and other system data. This architecture ensures modularity, scalability, security, efficient data management, and seamless communication between the user interface and the database.

The platform follows a Three-Tier Architecture:

- Presentation Layer (React.js)
- Business Logic Layer (Node.js + Express.js)
- Database Layer (PostgreSQL)

Functional Requirements

- User Authentication
- Student Profile Management
- Internal Project Submission
- External Project Submission
- Idea Showcase
- Faculty Assignment
- Faculty Review
- Credit Point Allocation
- Repository Access Control
- Student Networking
- Notifications
- Project Collaboration
- Likes, Comments, Views
- Leaderboard
- Admin Dashboard

Non-Functional Requirements

Performance

The system should provide fast and efficient performance to ensure a smooth user experience.

- Response time should be less than 3 seconds for normal operations.
- Support multiple concurrent users without performance degradation.
- Enable fast project search and retrieval.
- Handle large numbers of project uploads efficiently.

Security

The system should ensure secure access and protect user data from unauthorized access.

- JWT-based user authentication.
- Password encryption using secure hashing algorithms (bcrypt).
- Role-Based Access Control (Student, Faculty, Admin).
- Secure file upload validation.
- HTTPS-based secure communication.

Reliability

The platform should provide consistent and dependable operation.

- Daily database backup.
- Automatic error handling and recovery.
- Maintain data integrity during transactions.
- Ensure high system availability with minimal downtime.

Scalability

The application should support future growth without affecting system performance.

- Support increasing numbers of users.
- Allow addition of new departments and courses.
- Easily accommodate more projects and repository data.
- Modular architecture for future feature enhancements.

Usability

The system should be simple, user-friendly, and accessible to all users.

- Responsive user interface for different screen sizes.
- Easy navigation with intuitive menus.
- Simple project submission and approval workflow.
- Consistent user interface across all modules.

Maintainability

The software should be easy to update, debug, and maintain.

- Modular and well-structured codebase.
- Well-documented APIs.
- Easy bug fixing and feature enhancement.
- Reusable components for faster development.

Compatibility

The application should work consistently across different platforms and browsers.

- Compatible with Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
- Support Windows, Linux, and macOS operating systems.
- Responsive across desktops, laptops, tablets, and mobile devices.

Availability

The platform should remain accessible whenever users need it.

- Available 24×7 for students, faculty, and administrators.
- Ensure uninterrupted access during normal academic activities.
- Minimize downtime through regular monitoring and maintenance.

File Management

The system should securely manage uploaded project files.

- Support uploading of PDF, PPT, GitHub links, demo videos, and Vercel links.
- Validate file formats and file sizes before upload.
- Store uploaded files securely.
- Prevent unauthorized file access.

Database Management

The database should ensure efficient storage and retrieval of information.

- Use PostgreSQL as the relational database.
- Maintain data consistency using constraints and relationships.
- Optimize queries for faster data retrieval.
- Support backup and recovery mechanisms.

Database Tables

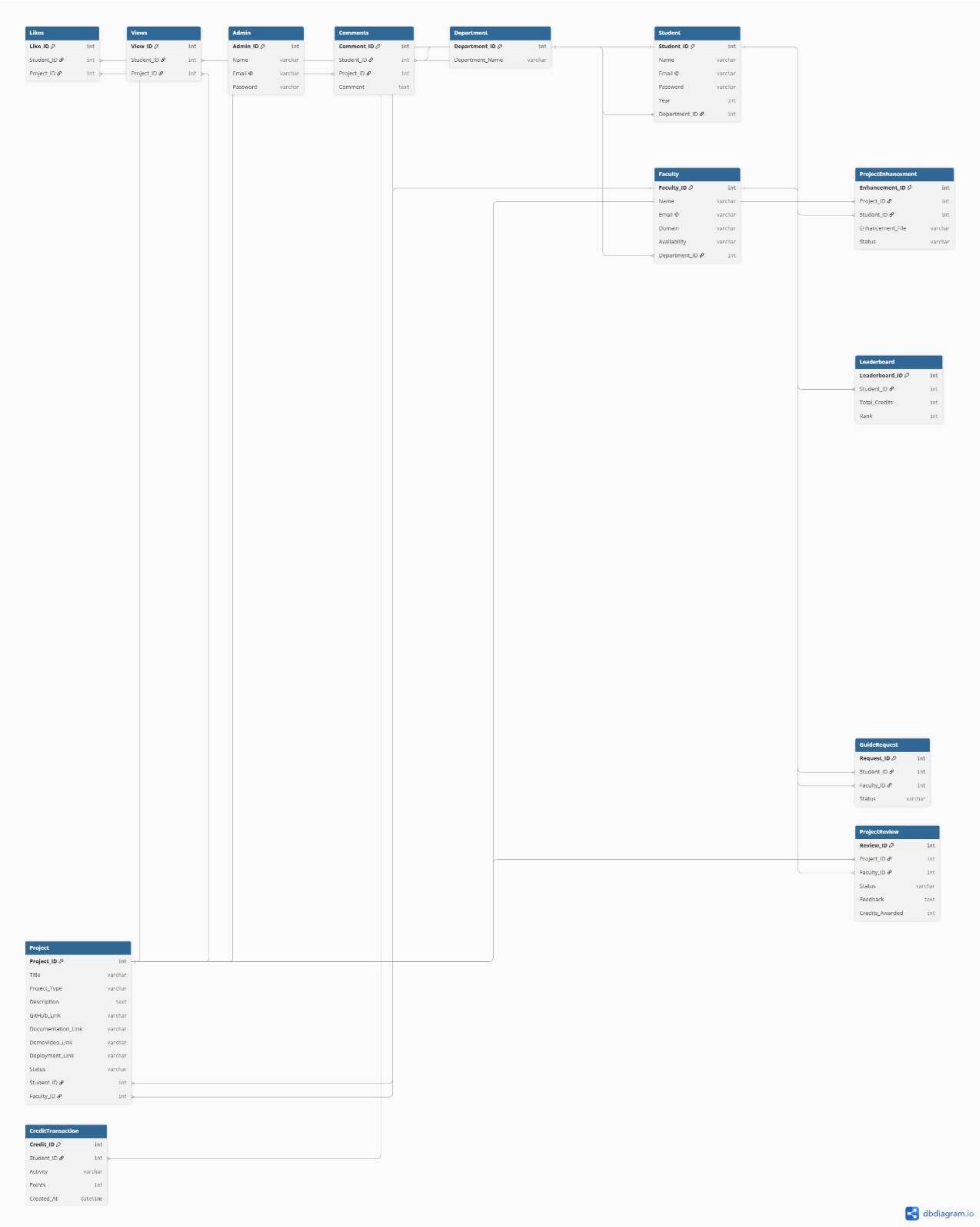
- Users
- Students
- Faculty
- Administrator
- Departments
- Faculty Availability
- External Guide Requests
- Internal Projects
- External Projects
- Idea Showcase
- Project Files
- Project Reviews
- Collaboration Requests
- Credit Points
- Leaderboard
- Likes
- Comments
- Views
- Notifications

Expected Outcomes

- A centralized digital repository for student projects.
- Transparent faculty-based project approval.
- Automatic reviewer assignment for internal projects.
- Faculty-guided external project showcase.

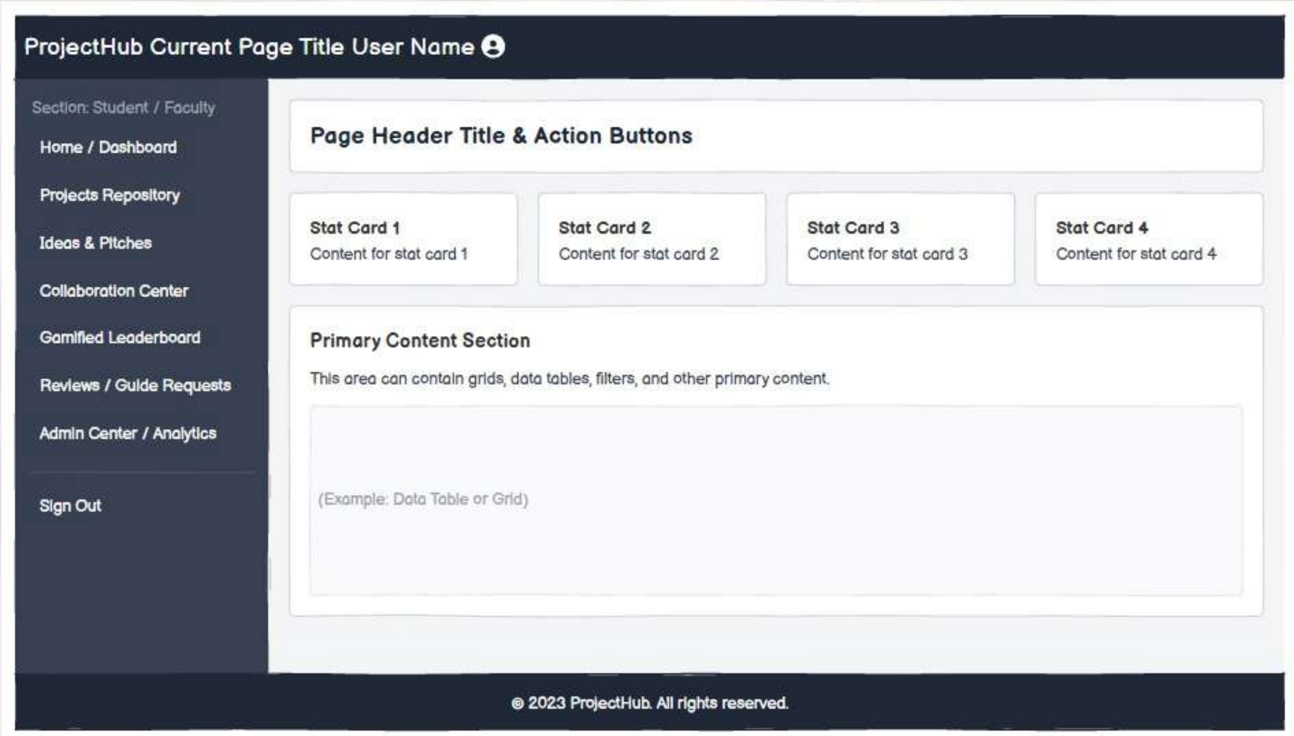
- Promotion of innovative ideas through the Idea Showcase.
- Fair contribution-based Credit Point system.
- Secure access to repositories based on student contributions.
- Increased collaboration through project enhancement.
- Healthy competition through student leaderboards.
- Better visibility of projects using likes, comments, and views.

ER Diagram



Likes
Like_ID int
Student_ID int
Project_ID int
Views
View_ID int
Student_ID int
Project_ID int
Admin
Admin_ID int
Name varchar
Email varchar
Password varchar
Comments
Comment_ID int
Student_ID int
Project_ID int
Comment text
Department
Department_ID int
Department_Name varchar
Student
Student_ID int
Name varchar
Email varchar
Password varchar
Year int
Department_ID int
Faculty
Faculty_ID int
Name varchar
Email varchar
Domain varchar
Availability varchar
Department_ID int
ProjectEnhancement
Enhancement_ID int
Project_ID int
Student_ID int
Enhancement_File varchar
Status varchar
Leaderboard
Leaderboard_ID int
Student_ID int
Total_Credits int
Rank int
GuideRequest
Request_ID int
Student_ID int
Faculty_ID int
Status varchar
ProjectReview
Review_ID int
Project_ID int
Faculty_ID int
Status varchar
Feedback text
Credits_Awarded int
Project
Project_ID int
Title varchar
Project_Type varchar
Description text
GitHub_Link varchar
Documentation_Link varchar
DemoVideo_Link varchar
Deployment_Link varchar
Status varchar
Student_ID int
Faculty_ID int
CreditTransaction
Credit_ID int
Student_ID int
Activity varchar
Points int
Created_At datetime
dbdiagram.io

UI Sketch


ProjectHub Current Page Title User Name
Section: Student / Faculty
Home / Dashboard
Projects Repository
Ideas & Pitches
Collaboration Center
Gamified Leaderboard
Reviews / Guide Requests
Admin Center / Analytics
Sign Out
Page Header Title & Action Buttons
Stat Card 1
Content for stat card 1
Stat Card 2
Content for stat card 2
Stat Card 3
Content for stat card 3
Stat Card 4
Content for stat card 4
Primary Content Section
This area can contain grids, data tables, filters, and other primary content.
(Example: Data Table or Grid)
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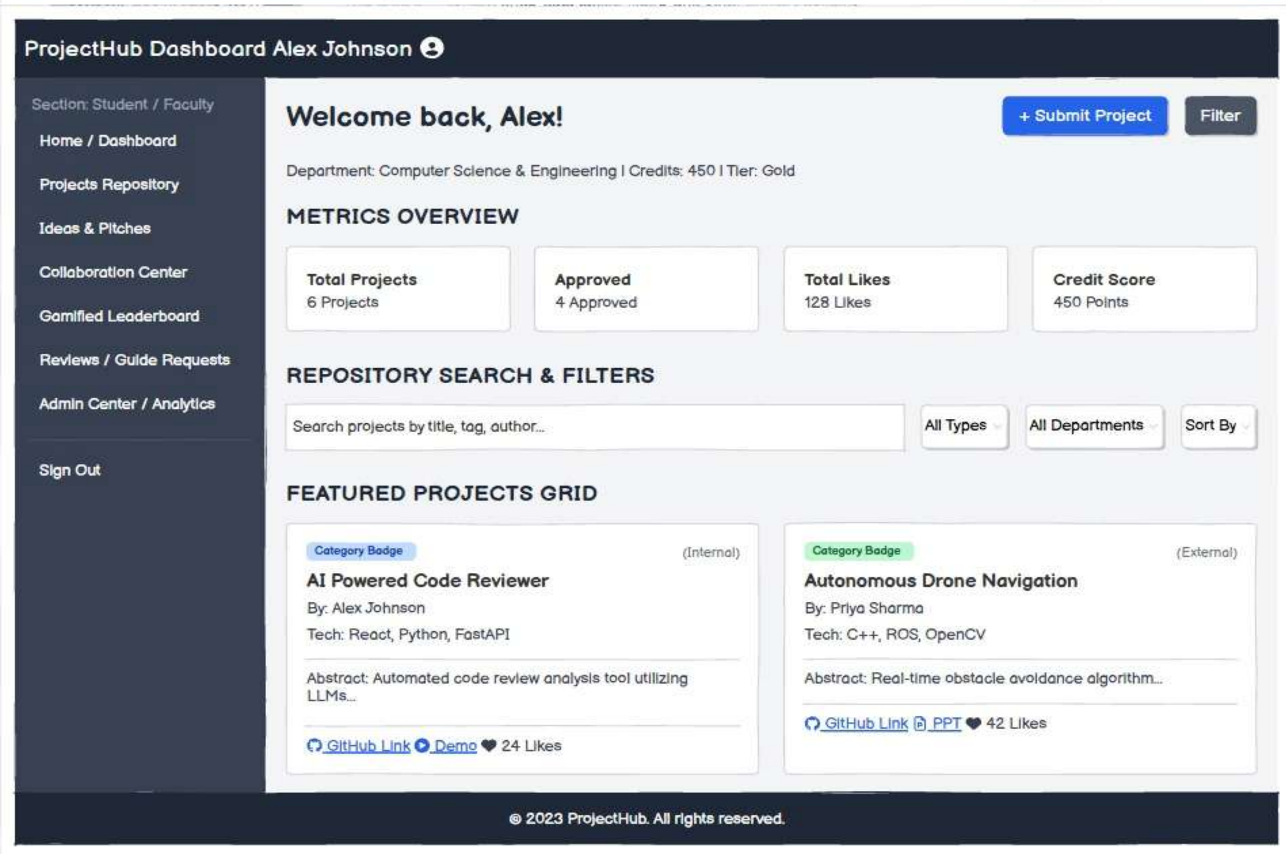
ProjectHub Dashboard Alex Johnson
Section: Student / Faculty
Home / Dashboard
Projects Repository
Ideas & Pitches
Collaboration Center
Gamified Leaderboard
Reviews / Guide Requests
Admin Center / Analytics
Sign Out
Welcome back, Alex!
+ Submit Project
Filter
Department: Computer Science & Engineering | Credits: 450 | Tier: Gold
METRICS OVERVIEW
Total Projects
6 Projects
Approved
4 Approved
Total Likes
128 Likes
Credit Score
450 Points
REPOSITORY SEARCH & FILTERS
Search projects by title, tag, author...
All Types
All Departments
Sort By
FEATURED PROJECTS GRID
Category Badge
(Internal)
AI Powered Code Reviewer
By: Alex Johnson
Tech: React, Python, FastAPI
Abstract: Automated code review analysis tool utilizing LLMs...
GitHub Link
Demo
24 Likes
Category Badge
(External)
Autonomous Drone Navigation
By: Priya Sharma
Tech: C++, ROS, OpenCV
Abstract: Real-time obstacle avoidance algorithm...
GitHub Link
PPT
42 Likes
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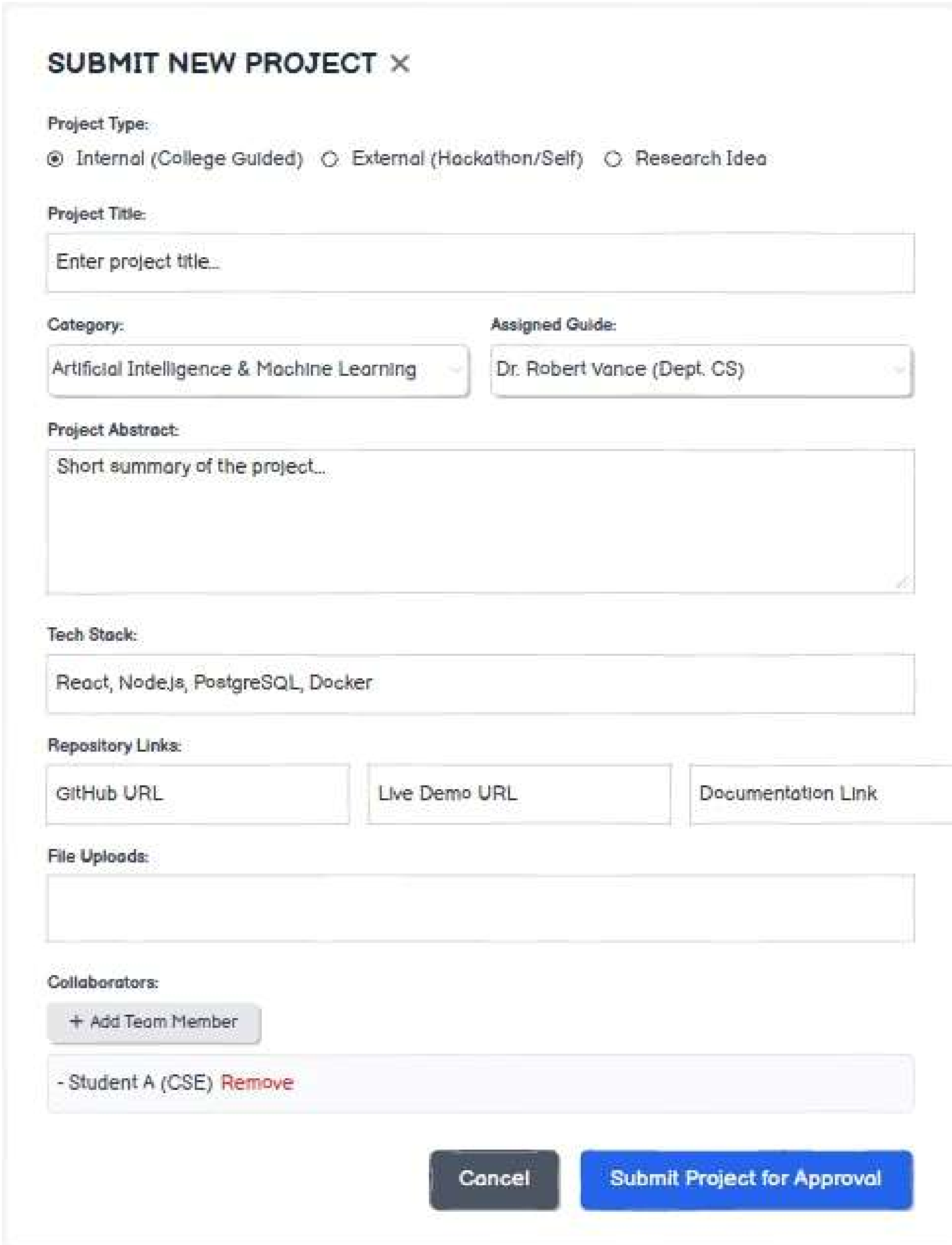
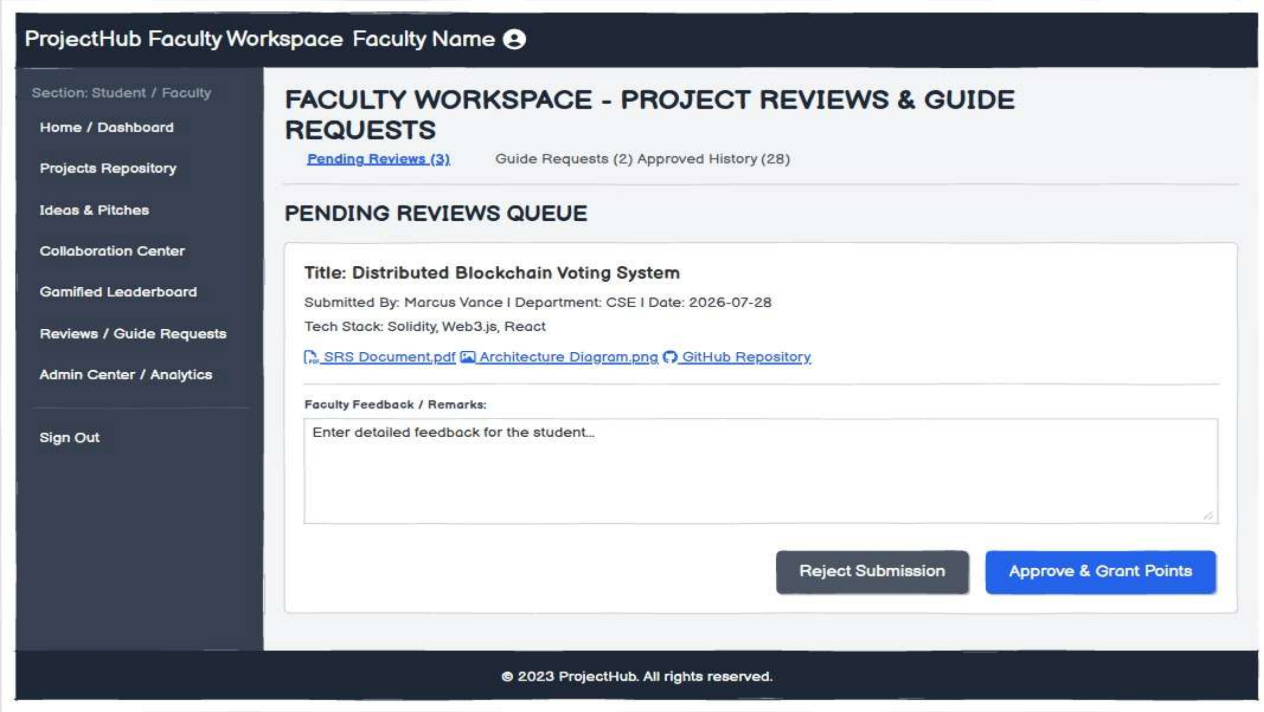

SUBMIT NEW PROJECT
Project Type:
Internal (College Guided)
External (Hackathon/Self)
Research Idea
Project Title:
Enter project title...
Category:
Artificial Intelligence & Machine Learning
Assigned Guide:
Dr. Robert Vance (Dept. CS)
Project Abstract:
Short summary of the project...
Tech Stack:
React, Node.js, PostgreSQL, Docker
Repository Links:
GitHub URL
Live Demo URL
Documentation Link
File Uploads:
Collaborators:
+ Add Team Member
- Student A (CSE) Remove
Cancel
Submit Project for Approval


ProjectHub Faculty Workspace Faculty Name
Section: Student / Faculty
Home / Dashboard
Projects Repository
Ideas & Pitches
Collaboration Center
Gamified Leaderboard
Reviews / Guide Requests
Admin Center / Analytics
Sign Out
FACULTY WORKSPACE - PROJECT REVIEWS & GUIDE REQUESTS
Pending Reviews (3)
Guide Requests (2) Approved History (28)
PENDING REVIEWS QUEUE
Title: Distributed Blockchain Voting System
Submitted By: Marcus Vance I Department: CSE I Date: 2026-07-28
Tech Stack: Solidity, Web3.js, React
SRS Document.pdf
Architecture Diagram.png
GitHub Repository
Faculty Feedback / Remarks:
Enter detailed feedback for the student...
Reject Submission
Approve & Grant Points
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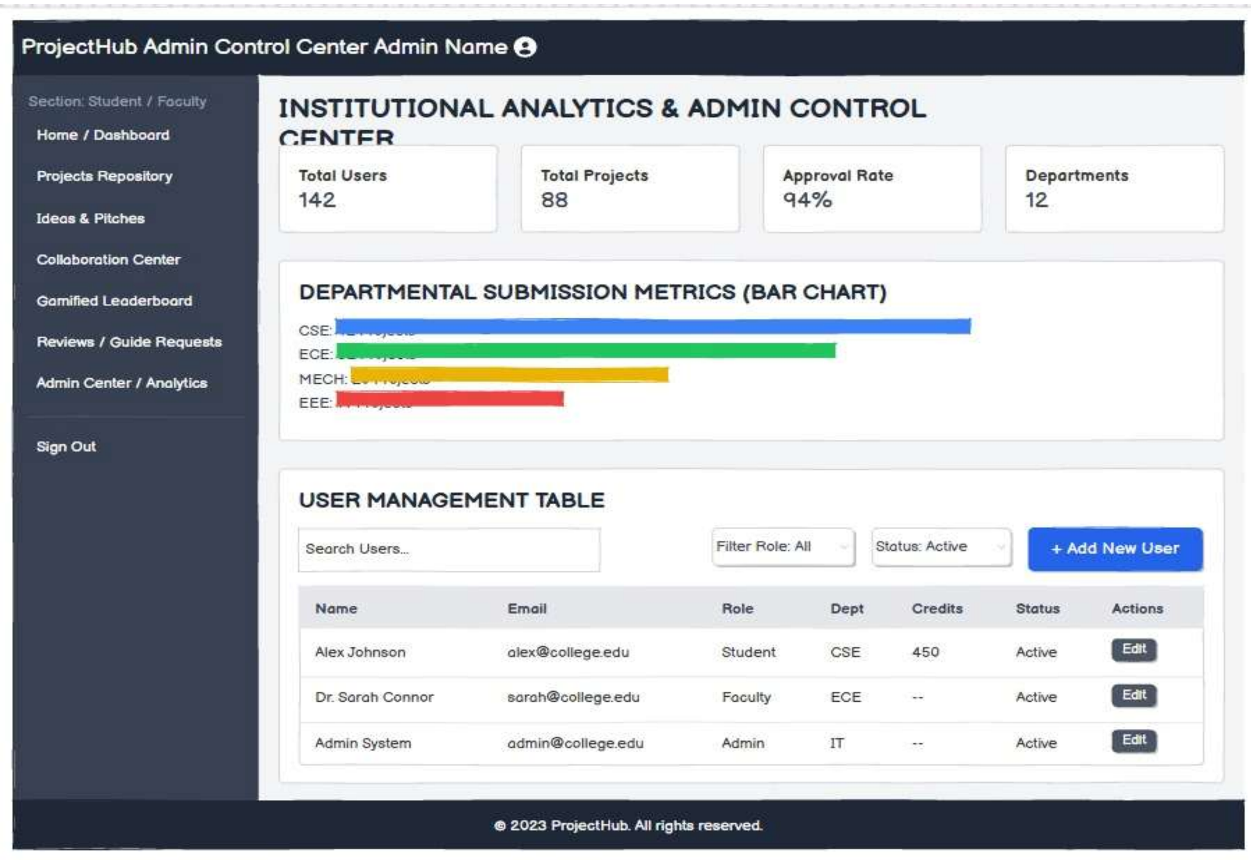
ProjectHub Admin Control Center Admin Name
Section: Student / Faculty
Home / Dashboard
Projects Repository
Ideas & Pitches
Collaboration Center
Gamified Leaderboard
Reviews / Guide Requests
Admin Center / Analytics
Sign Out
INSTITUTIONAL ANALYTICS & ADMIN CONTROL CENTER
Total Users
142
Total Projects
88
Approval Rate
94%
Departments
12
DEPARTMENTAL SUBMISSION METRICS (BAR CHART)
CSE:
ECE:
MECH:
EEE:
USER MANAGEMENT TABLE
Search Users...
Filter Role: All
Status: Active
+ Add New User
Name
Email
Role
Dept
Credits
Status
Actions
Alex Johnson
alex@college.edu
Student
CSE
450
Active
Edit
Dr. Sarah Connor
sarah@college.edu
Faculty
ECE
--
Active
Edit
Admin System
admin@college.edu
Admin
IT
--
Active
Edit
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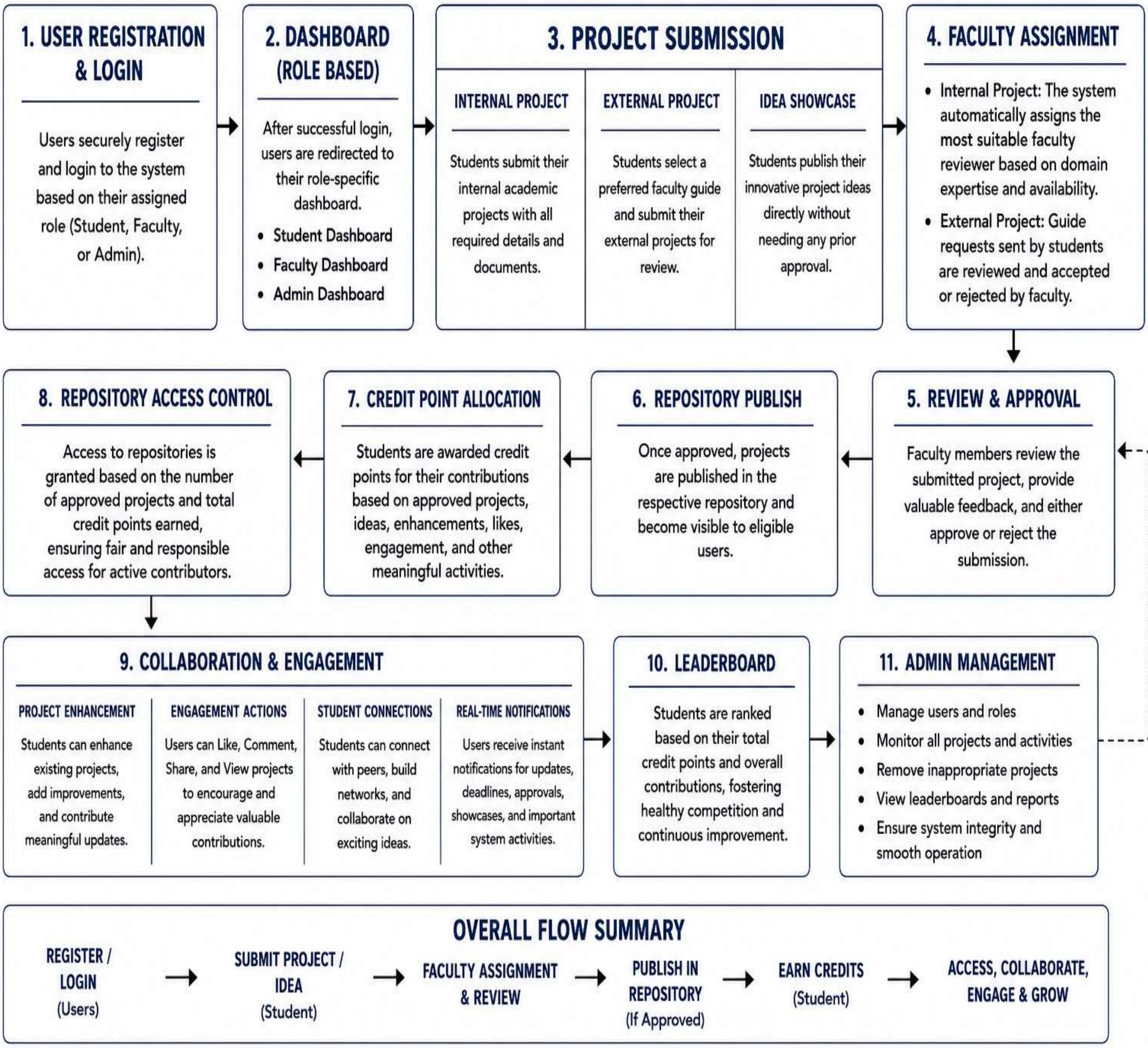
System Workflow



Use Cases

Actors

- Student
- Faculty
- Administrator

Use Cases

- Login
- Upload Project
- Submit Idea
- Review Project
- Approve Project
- Award Credits
- Connect Students
- View Repository
- View Leaderboard
- Remove Projects

Use Case Description

Each use case describes how a user interacts with the system.

Example:

Use Case: Internal Project Submission

Actor: Student

Precondition: Student Login

Flow:

- Student uploads project.
- System assigns faculty.
- Faculty reviews project.
- Faculty approves project.
- Credits are awarded.
- Project is published.

Future Enhancements

1. Mobile Application Support
2. AI-Based Project Recommendation
3. Multi-College Repository Integration
4. Cloud File Storage
5. Email & Push Notification Service
6. QR Code Generation for Project Showcase
7. Project Version Control
8. Project Bookmarking
9. Project Plagiarism Detection
10. Analytics Dashboard
11. Faculty Performance Reports
12. Department-Wise Performance Analysis
13. Certificate Generation for Top Contributors
14. Internship & Placement Integration
15. Industry Mentor Collaboration

References

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2. Node.js Official Documentation – https://nodejs.org
3. Express.js Documentation – https://expressjs.com
4. PostgreSQL Documentation – https://www.postgresql.org/docs/
5. JWT Documentation – https://jwt.io
6. GitHub Documentation – https://docs.github.com
7. IEEE 830 Software Requirements Specification Standard
8. PostgreSQL Official Manual
9. React Router Documentation
10. Bootstrap Documentation